

TAM THUC STRATEGEM

Full Project Structure and Codebase Guide v1.0

1. RECOMMENDED PROJECT STRUCTURE

```
tamthuc-strategem/  
├── backend/  
│   ├── app/  
│   │   ├── api/ (routers)  
│   │   ├── core/ (config, security)  
│   │   ├── engines/ (qimen, liuren, taiyi)  
│   │   ├── services/ (rule_engine, rag_service, report_generator)  
│   │   ├── models/ (pydantic + sqlalchemy)  
│   │   └── main.py  
│   ├── tests/  
│   └── requirements.txt  
├── frontend/  
│   ├── src/  
│   │   ├── components/ (UI components)  
│   │   ├── pages/ (Dashboard, Query, Report...)  
│   │   ├── hooks/  
│   │   ├── lib/ (api client, utils)  
│   │   └── app/ (Next.js app router)  
│   ├── package.json  
├── shared/ (types, constants)  
└── docs/ (tất cả PDF đã tạo)
```

2. CORE MODULES CẦN IMPLEMENT

1. ****Calculation Engines**** (ưu tiên Kỳ Môn trước)
2. ****Rule Engine + Pattern Matching****
3. ****RAG Knowledge Service****
4. ****Report Generation Service****
5. ****Interactive Chart Component**** (SVG/Canvas)
6. ****Authentication & User Management****
7. ****Query History & Dashboard****

3. NEXT STEPS RECOMMENDED

1. Setup monorepo (TurboRepo hoặc Nx)
2. Implement QiMenEngine đầy đủ + test với 50+ classical examples
3. Build basic FastAPI + PostgreSQL + Redis
4. Develop core UI components + Interactive Chart
5. Integrate RAG (Chroma + OpenAI embedding)
6. Build end-to-end flow cho 1 use case (ví dụ: Timing Optimizer)
7. Add expert validation workflow